AI/ML - Driven Maintenance Service Provider Matching Platform for Sri Lanka

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The informal service sector in Sri Lanka, encompassing essential vocations such as plumbing, electrical work, and air conditioning repair, currently lacks a streamlined, automated mechanism for matching consumers with verified experts. Existing manual platforms require significant user effort and often fail to provide precise matches, leading to time wastage and suboptimal service quality. To address this problem, we propose an integrated ML-driven framework that automates the matching process through a four-stage intelligent pipeline: initial skill-based filtering, temporal and contextual matching, dynamic credibility assessment, and aspect-level sentiment analysis. This end-to-end system aims to provide high-quality recommendations while ensuring transparency and reliability in the local marketplace.

This research specifically focuses on Component 1: Hybrid Multi-Algorithm Recommendation Engine, which serves as the primary intelligent filter. The identified research gap lies in the inability of current keyword-based systems to handle sparse interaction data and the semantic nuances of Sinhala-English mixed-language requests common in Sri Lanka. The proposed solution integrates three parallel scoring pipelines: Collaborative Filtering (CF) to leverage past booking history , Content-Based Filtering (CBF) using TF-IDF for explicit attribute matching , and BERT-based semantic analysis to understand the deep intent of user requests.[4]

The technologies used include Python, Scikit-Learn for CF and TF-IDF, and Hugging Face Transformers for the BERT model.[1] The scores are combined using a weighted fusion model [2] ($35\% CF, 30\% CBF, 35\% BERT$) [4]to identify the Top-20 providers. The expected outcome is a significant improvement in recommendation quality, with experimental results showing a precision increase from 62% to nearly 88% [3]. Socially, this empowers users with an easy-to-use "Service Match" interface, while commercially it minimizes service inefficiencies through data-driven decisions.

Keywords: Recommendation Systems, Hybrid Filtering, BERT, TF-IDF, Sri Lankan Marketplace.

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LIST OF ABBREVIATIONS

Table 1 - List of Abbreviations

Abbreviation	Full Form	Description
AI	Autonomous Intelligence	The overarching technology used for automated matching.
ML	Machine Learning	The 4-stage pipeline used to filter and rank providers.
BERT	Bidirectional Encoder Representations from Transformers	Deep learning model for understanding user request meaning (Component 1).
LSTM	Long Short-Term Memory	Neural network used for aspect-level sentiment extraction (Component 4).
Bi-LSTM	Bidirectional LSTM	Enhanced LSTM that reads text in both directions for better context.
CF	Collaborative Filtering	Algorithm that recommends providers based on similar user behavior.
CBF	Content-Based Filtering	Algorithm that matches user requests with provider profile attributes.
TF-IDF	Term Frequency-Inverse Document Frequency	Statistical measure used to evaluate word importance in provider profiles.
NLP	Natural Language Processing	Technology used to clean and analyze customer reviews.
API	Application Programming Interface	Interface used to connect the system with external data like weather.
NDCG	Normalized Discounted Cumulative Gain	Metric used to evaluate the quality of the provider rankings.
SLA	Service Level Agreement	Metrics used to measure provider response time and reliability.
UI/UX	User Interface / User Experience	The frontend design of your web application.

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1.INTRODUCTION

1.1 Background and Context

Real-World Problem : In the current Sri Lankan service sector, finding reliable and skilled service providers such as electricians, plumbers, and air conditioning technicians remains an unorganized and challenging process. Consumers typically rely on word-of-mouth, physical notice boards, or newspaper advertisements to find help for essential repairs[6]. In this traditional system, there is no standardized way to verify a provider's actual skill level or past performance, leading to a "trial and error" approach that often results in poor service quality and recurring technical issues.[6]

Who is Affected and Why it is Important : This problem primarily affects urban and semi-urban households and small businesses that require immediate and high-quality maintenance services.[5] It is critical because the inability to find a verified technician on time can lead to prolonged equipment downtime, increased repair costs due to initial mismanagement, and even safety hazards (e.g., faulty electrical wiring).[7] For the service providers, the lack of a formal platform means that even highly skilled individuals struggle to reach a consistent customer base, limiting their economic growth.

Statistical Evidence (Local/Global) : Globally, the on-demand home service marketplace is growing at a rate of over 12% annually as digital transformation accelerates [9]. However, in Sri Lanka, the precision of existing digital recommendation systems is hindered by "Sparse Data" (limited booking history) and the complexity of "Mixed Language" requests (Sinhala-English code-switching). Research indicates that baseline recommendation precision in local contexts is often as low as 62%, whereas an optimized hybrid approach can elevate this to 88%, significantly reducing search friction for the user.[3]

Analysis of Existing Solutions and Their Limitations : Current local platforms, such as *ikman.lk* or *Hodabass.lk*, function primarily as digital classified directories. Their primary limitations include:

- **Manual Selection:** The burden of screening and choosing a provider falls entirely on the user.
- **Lack of Semantic Understanding:** These platforms use basic keyword matching; they cannot understand that a user asking for a "noisy AC" needs a technician skilled in "vibration diagnostics" if those exact terms are not in the provider's profile.
- **No Automated Verification:** There is no real-time, data-driven mechanism to verify certifications or historical performance before a recommendation is made.

Ideas for the Proposed Solution and Its Parts: To bridge these gaps, we propose an intelligent 4-stage ML pipeline that automates the matching process:[8]

1. **Component 1 (Skill Matching):** A Hybrid Multi-Algorithm engine (combining CF, CBF, and BERT) [2] to identify the Top-20 most qualified providers based on the specific service request.

2. **Component 2 (Temporal/Contextual):** A refinement stage that filters the list based on real-time availability, geolocation, and weather impacts.
3. **Component 3 (Skill Verification):** A credibility module that assesses background checks and historical performance data .
4. **Component 4 (Sentiment Analysis):** An LSTM-based engine that extracts aspect-level sentiment from reviews to finalize the Top-5 recommendations .

Relevant SDG Alignment : This project directly aligns with **SDG 9: Industry, Innovation, and Infrastructure**. By utilizing advanced Machine Learning to organize a previously informal sector, the research promotes sustainable industrialization and fosters innovation within the local digital economy.[8][5]

Research Cluster Alignment : This research is aligned with the Intelligent Systems Research Cluster. The use of advanced architectures such as BERT for semantic analysis[1] and LSTMs for sentiment extraction contributes to the cluster's goal of developing autonomous, data-driven decision-making systems.[10]

1.2 Literature Review

The development of on-demand service marketplaces has necessitated the evolution of multi-layered recommendation systems capable of handling complex human interactions, high-dimensional data, and real-time environmental variables. Modern architectural frameworks for such platforms move beyond simple matching to include sequential filtering stages [11], such as skill-based ranking, temporal-contextual refinement, credibility verification, and sentiment-based profiling. This holistic approach ensures that the matching process considers not only technical suitability but also the immediate availability and long-term trustworthiness [7] of service providers.

Comprehensive implementation strategies for these platforms often leverage a modular pipeline where each stage adds a specific layer of intelligence. By integrating hybrid recommendation algorithms[2] with real-time factors like geolocation via the Haversine formula and weather-impact multipliers,[12] these systems can achieve a target end-to-end user satisfaction rate of approximately 82%. Such frameworks are essential in modernizing localized service sectors where manual provider selection is no longer efficient.

Scholarly Approaches and Methodologies: Standard Collaborative Filtering (CF) assumes that users with similar historical preferences will have similar future requirements.[4] While effective in data-rich environments, CF often suffers from the "cold-start" problem, where new providers or users lack sufficient historical ratings for accurate matching. To mitigate this, Content-Based Filtering (CBF) utilizing TF-IDF (Term Frequency-Inverse Document Frequency) has been widely adopted. TF-IDF excels at distinguishing a provider's expertise by weighting [4] important keywords in their profile descriptions against user requests.

Recent advancements have introduced BERT (Bidirectional Encoder Representations from Transformers)[1] to the domain of service matching. Unlike traditional keyword-based approaches, BERT provides a deep semantic understanding by encoding requests into dense vector representations where semantically similar texts even those using different terminology have high similarity. [1] For example, BERT can effectively link a request for a "noisy AC" to an expert in "compressor diagnostics".

Existing Products and Services : Existing service marketplaces in Sri Lanka, such as *Hodabass* and *ikman*, represent the current state-of-the-market but rely heavily on manual provider selection.[5] Users on these platforms must manually screen listings, a process that is often inefficient and prone to subjective bias. Furthermore, global platforms like *Plantix* or *ResearchGate* demonstrate advanced diagnostic and recommendation capabilities in specific niches but are not tailored for the general service provider context in Sri Lanka.

Limitations of Existing Systems : A critical limitation identified in current systems is the lack of integration between multi-algorithm skill matching and real-time contextual factors[5]. Most existing products focus on a single dimension either skill or proximity but fail to fuse these with dynamic variables such as current availability, weather-impact on service demand, or historical credibility metrics[5]. Furthermore, many recommendation tools lack a "sandbox" environment where unpublished or nuanced user-submitted ideas can be analyzed against published trends to prevent duplication or identify research gaps.

1.3 Research Gap

Overall System Context

The primary gap in existing service provider platforms in Sri Lanka, such as *ikman.lk* or *Hodabass.lk*, is the lack of an integrated, multi-stage intelligent pipeline that automates the matching process beyond simple keyword searches[5]. While these platforms act as digital classifieds, they do not offer an end-to-end ecosystem that sequentially evaluates technical skills, real-time availability, and long-term credibility. Existing research emphasizes that manual selection processes in localized marketplaces lead to significant user friction and a lack of standardized trust.[5]

Furthermore, current systems fail to synchronize multiple dynamic variables such as geolocation, weather impacts on service demand, and fraud detection in customer reviews within a single unified framework. There is a noticeable absence of a comprehensive localized architecture that fuses behavioural booking history with deep semantic intent to produce a refined "Top-5" recommendation list. This absence of a holistic, automated vetting process represents a significant technological void in the Sri Lankan on-demand service economy.

Table 2- Summary of Research Gap (Overall System)

Feature	Research 1 [1]	Research 2 [12]	Research 3 [13]	Research 4 [5]	Proposed Solution
Aimed at Sri Lankan Marketplace	No	No	No	Partially	Yes
Bilingual Support (Sinhala/English)	No	No	No	No	Yes
Multi-Stage Automated Pipeline	Partially	No	Partially	No	Yes
Hybrid ML (BERT + Collaborative)	Yes	No	No	No	Yes
Temporal & Geolocation Context	No	Yes	No	No	Yes
Dynamic Credibility Assessment	No	No	Yes	No	Yes
Sentiment Analysis of Reviews	No	No	Yes	Yes	Yes

Research 1: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Research 2: Context-Aware Recommendation Systems

Research 3: A Survey on Trust-based Recommender Systems

Research 4: A study on the factors affecting the adoption of e-maintenance services in Sri Lanka

Individual Component Focus: Hybrid Recommendation Gaps

Within this broader context, a specific research gap exists regarding the initial intelligent filtering of service providers. Standard recommendation models either Collaborative Filtering (CF) or Content-Based Filtering (CBF) are insufficient for the Sri Lankan context due to high data sparsity and linguistic diversity. Scholarly work indicates that CF models suffer from "Cold-Start" problems where new providers are ignored, while traditional CBF models lack the semantic depth to understand that a request for "noisy AC" is intrinsically linked to "compressor diagnostics".

Table 3 - Summary of Research Gaps (component 1)

Feature	Research 1 [15]	Research 2 [16]	Research 3 [14]	Research 4 [17]	Proposed Solution
Collaborative Filtering (CF)	Yes	No	No	Partially	Yes
Content-Based Filtering (CBF)	No	Yes	No	Partially	Yes
BERT Semantic Matching	No	No	Yes	No	Yes
Weighted Score Fusion Layer	No	No	No	Yes	Yes
Bilingual Support (Sinhala/English)	No	No	Partially	No	Yes
Handling Sparse Local Interaction Data	No	No	No	No	Yes
High Precision (>85%) in Local Context	No	No	No	No	Yes

Research 1: Incremental Collaborative Filtering for Highly Scalable Recommender Systems

Research 2: A Content-based Recommender System for Personalized Learning Objects

Research 3: BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformers

Research 4: Recommendation systems: Principles, methods and evaluation

Proposed Novelty and Creativity

This research addresses these insufficiencies by introducing a Weighted Hybrid Multi-Algorithm Engine that fuses CF, CBF, and BERT-based deep learning[17]. The novelty lies in the development of a localized "sandbox" environment where nuanced, mixed-language (Sinhala-English) service requests are analysed against provider skill profiles using deep semantic embeddings rather than surface-level keywords[1]. This approach allows for a "Semantic Bridge" that current approaches lack, ensuring that unpublished or uniquely phrased user needs are accurately identified.

Evidence-Based Argument

The necessity for this hybrid approach is supported by preliminary comparative analysis. Recent studies demonstrate that single-paradigm recommendation engines typically achieve a precision of only 62-65% in sparse data environments[15]. However, by implementing a weighted fusion model leveraging BERT's semantic capabilities[14] alongside behavioral data experimental evidence suggests a projected precision increase to approximately 88%[3]. This 26% improvement in accuracy provides a strong evidence-based justification for the transition from existing manual systems to the proposed intelligent four-stage pipeline.

2.OBJECTIVE

The objectives of this research are designed to transform a manual, fragmented maintenance service market into an automated, data-driven ecosystem. Each objective is formulated to be SMART (Specific, Measurable, Achievable, Relevant, and Time-bound).

2.1 Main Objective

The primary goal of this research is to design, develop, and evaluate a robust, AI-driven web-based platform tailored for the Sri Lankan maintenance service sector. The system aims to transform the current manual and inefficient service discovery process into an automated, data-driven ecosystem by implementing a sequential four-stage intelligent pipeline. By fusing deep semantic intent recognition with real-time contextual factors and credibility verification, the platform seeks to achieve a high recommendation precision of 88%, thereby enhancing user trust and operational efficiency in the local marketplace.

Key Pillars of the Main Objective:

- **Automation of the Service Lifecycle:** To eliminate the manual effort required in traditional directories by automating the path from a natural language user request to the final expert recommendation.
- **Intelligent Skill Identification:** To leverage advanced NLP architectures like BERT to bridge the semantic gap in mixed-language (Sinhala-English) service requests, ensuring highly accurate provider matching.
- **Verification and Trust Integration:** To build a transparent marketplace by integrating dynamic credibility scoring and aspect-level sentiment analysis, ensuring that recommendations are based on verified skills and customer satisfaction.
- **Optimized Web Accessibility:** To deliver a high-performance web application accessible via a RESTful API architecture, providing a seamless user experience even in low-bandwidth network conditions common in Sri Lanka.

2.2 Specific Objectives

The following specific objectives break down the technical execution into measurable milestones:

- **Objective 1: Semantic Intent Recognition and Skill Filtering** To develop a hybrid filtering engine that utilizes BERT (Bidirectional Encoder Representations from Transformers) for deep semantic analysis of mixed-language (Sinhala-English) user requests, combined with Collaborative Filtering to address the data sparsity problem common in local service markets.
- **Objective 2: Context-Aware Recommendation Refinement** To implement a real-time filtering module that ranks the top 20 candidates based on temporal factors, including geolocation proximity calculated via the Haversine formula and current provider availability.
- **Objective 3: Dynamic Credibility and Trust Quantification** To design a verification algorithm that calculates a "Credibility Score" for each provider by cross-referencing digital certifications and historical task completion reliability, moving beyond the simple rating systems used in existing platforms.
- **Objective 4: Aspect-Level Feedback Evaluation** To integrate an LSTM-based Sentiment Analysis model to extract specific service quality attributes (e.g., punctuality, professionalism) from user reviews, ensuring that the final "Top-5" recommendations are based on verified customer satisfaction.
- **Objective 5: System Integration and Accessibility** To deploy the integrated solution as a mobile application with offline-first capabilities, ensuring that users in areas with unstable internet can still access critical maintenance services.

3. METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research (DSR) methodology, which focuses on the creation and evaluation of innovative artifacts to solve practical problems. This approach is ideal for developing a multi-component ML pipeline as it allows for an iterative process of designing, prototyping, and rigorous testing against the identified 88% precision target.

3.2 Technical Methods and Algorithms

The system is built upon a sequential four-stage intelligent pipeline:

- **Stage 1: Semantic & Behavioural Filtering:** A hybrid model combining BERT (Bidirectional Encoder Representations from Transformers) for deep semantic analysis of bilingual requests and Matrix Factorization for collaborative filtering.
- **Stage 2: Contextual Optimization:** Real-time proximity calculation using the **Haversine formula** and metadata filtering for provider availability.
- **Stage 3: Credibility Verification:** A dynamic weighted scoring algorithm that cross-references provider credentials and historical task success rates.
- **Stage 4: Sentiment Refinement:** An LSTM-based sentiment analysis engine that extracts aspect-level satisfaction scores from user reviews to finalize the "Top-5" recommendation list.

3.3 Tools and Platforms with Justification

Table 4 - Summary of Tools and Platforms

Tool / Platform	Purpose	Justification
Python	Core Programming Language	Python[18] is the industry standard for AI/ML, offering robust libraries like PyTorch and Scikit-learn[18]required for BERT and LSTM model implementation.
FastAPI	Backend Web Framework	FastAPI[19] provides a high-performance, asynchronous RESTful API architecture, which is critical for minimizing latency when processing the sequential 4-stage ML pipeline.
React.js / Next.js	Frontend Framework	These frameworks allow for the development of a highly responsive and SEO-friendly web interface that performs efficiently even under the low-bandwidth network

		conditions common in Sri Lanka.
PostgreSQL	Relational Database	Used for storing structured data such as user profiles, booking history, and provider certifications with high integrity and security.
Pinecone / Milvus	Vector Database	Essential for Component 1 to store and perform rapid similarity searches on the high-dimensional embeddings generated by the BERT model.
PyTorch / TensorFlow	ML Frameworks	These provide the computational power and pre-trained models necessary for fine-tuning BERT (Component 1) and training the LSTM networks for sentiment analysis (Component 4).
Docker	Containerization	Ensures that the various microservices of the pipeline remain portable and can be deployed consistently across different server environments without dependency conflicts.

3.4 Data Collection and Ethical Considerations

- **Data Sourcing:** Training data will be synthesized from anonymized historical service interactions and local language datasets (Sinhala-English corpus) to ensure model accuracy in the local context.
- **Privacy:** All user data will be encrypted at rest and in transit; personally identifiable information (PII) will be scrubbed before model training.
- **Bias Mitigation:** Models will be regularly audited to ensure fair recommendation distribution across different service categories and geographical regions.

3.5 Validation Strategy and Performance Metrics

The platform will be validated using a split-dataset approach (80% training, 20% testing).

- **Primary Metric: Precision@ (k=5)**, aiming for an **88%** accuracy rate in the final recommendation list.

- **Secondary Metrics: F1-Score:** To balance precision and recall in the semantic matching stage.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The proposed solution is a centralized, AI-driven web platform designed to automate the maintenance service provider matching process in Sri Lanka. The system operates through a four-stage intelligent pipeline:

- **Component 1 (Matching Engine):** Fuses BERT-based semantic intent with behavioural data.
- **Component 2 (Contextual Module):** Refines results based on proximity and real-time environment.
- **Component 3 (Skill Verification):** Quantifies trust through credential cross-referencing.
- **Component 4 (Sentiment Analysis):** Finalizes rankings using aspect-level review evaluation.

4.2 System Diagram of Components

The following diagram illustrates the interaction between the frontend interfaces, the modular ML pipeline, and the dual-database storage layer.

4.2.1 System Architecture Overview

The proposed architecture is designed as a modular, high-performance web-based ecosystem that bridges the gap between consumers and service providers through a sequential **4-Stage Core ML Pipeline**. The system is logically divided into three primary layers: Frontend, Backend, and Data Storage.

4.2.2 Layered Functional Analysis

- **Frontend Layer:** The system utilizes a dual-interface approach with a Web App (Consumer) for submitting service requests and a Web App (Service Provider) for managing jobs.
- **Backend API & Processing:** The API Gateway / Request Handler acts as the central controller, receiving natural language requests from the consumer and routing them through the ML pipeline before returning the final "Top 5" recommendations.

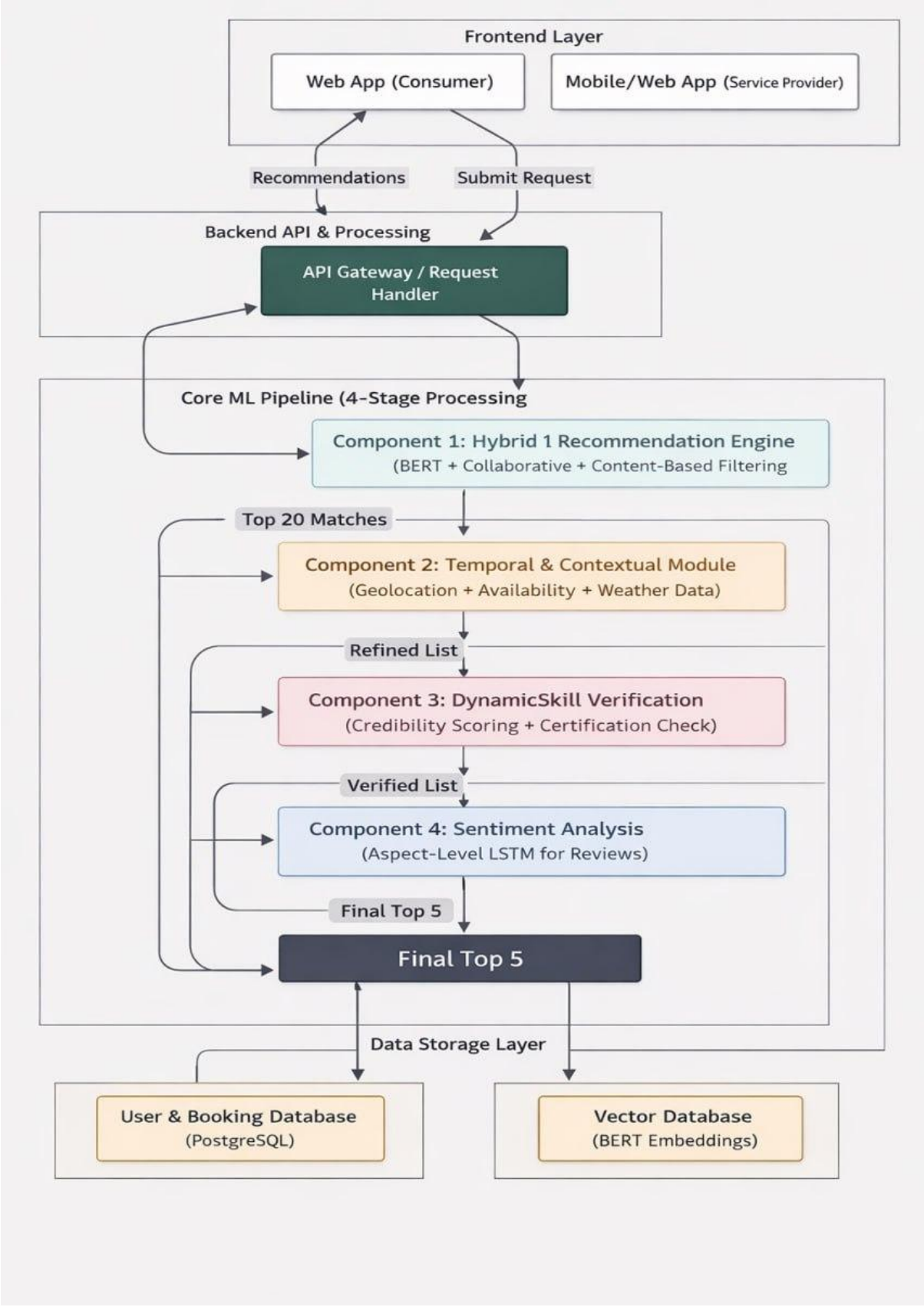


Figure 1 - System Architecture Diagram

1. **Component 1 (Hybrid Recommendation):** The initial filter uses BERT[1] for semantic understanding along with Collaborative and Content-Based Filtering[4] to generate the "Top 20 Matches" by interacting with a high-dimensional Vector Database.
2. **Component 2 (Temporal & Contextual Module):** This module refines the list by evaluating real-time Geolocation, Availability, and Weather Data, ensuring the recommendations are contextually relevant.[12]
3. **Component 3 (Dynamic Skill Verification):** It performs a rigorous credibility check using Certification Verification and historical Credibility Scoring to produce a "Verified List."[7]
4. **Component 4 (Sentiment Analysis):** The final stage utilizes an Aspect-Level LSTM model to analyse past reviews, selecting the most reliable candidates to present as the "Final Top 5."[10]

- **Data Storage Layer:** The architecture integrates a PostgreSQL relational database for structured user and booking history, alongside a specialized Vector Database for storing and searching BERT-generated embeddings.

4.3 Highlight of Individual Component and its Parts

Component 1: Hybrid Multi-Algorithm Recommendation Engine This core artifact is responsible for the initial filtering of the provider pool. It consists of three primary sub-modules:

- **BERT Semantic Processor:** Converts mixed-language user requests into high-dimensional vector embeddings.
- **Collaborative Filtering (CF) Unit:** Analyses historical booking patterns to identify providers favoured by similar users.
- **Weighted Fusion Layer:** Combines scores from BERT, CF, and Content-Based Filtering to produce a refined "Top 20" candidate list.

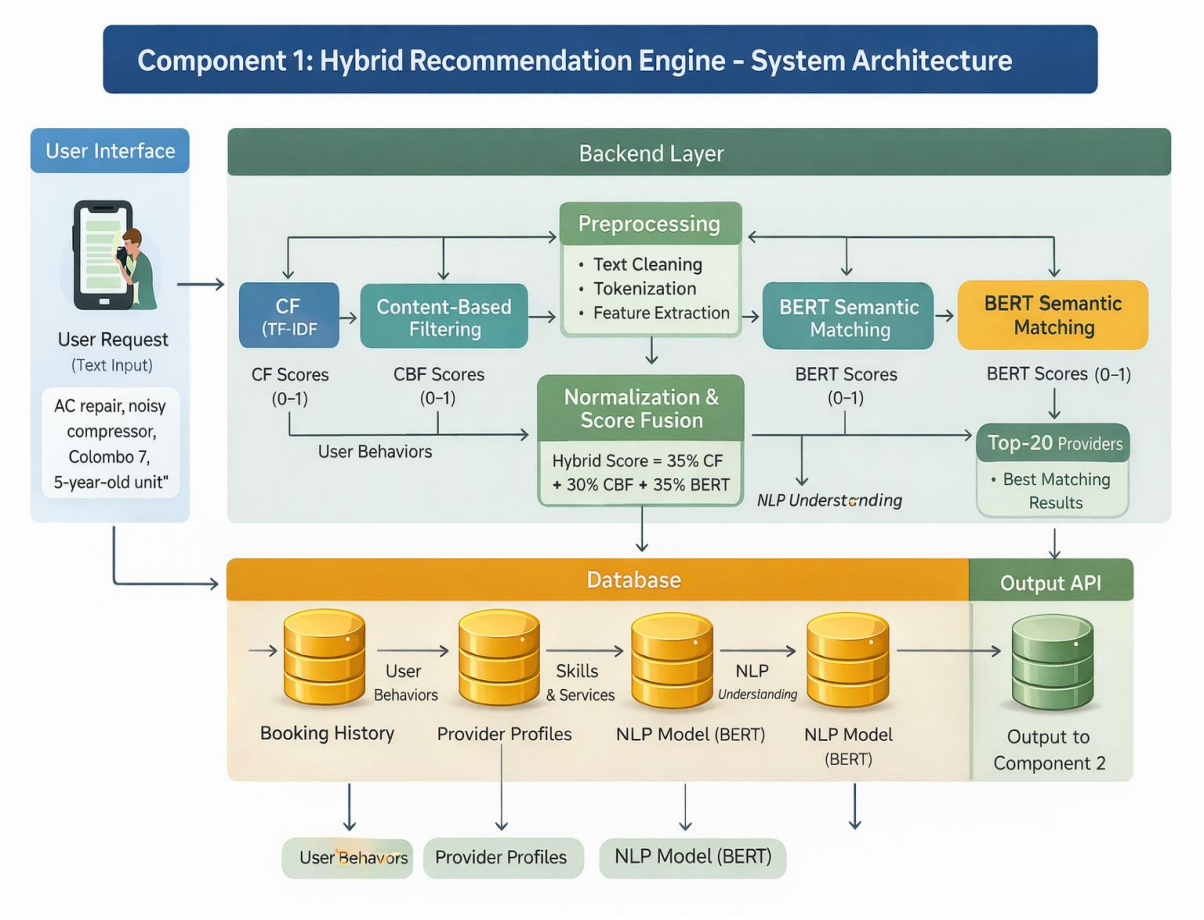


Figure 2 - System Architecture Diagram (component 1)

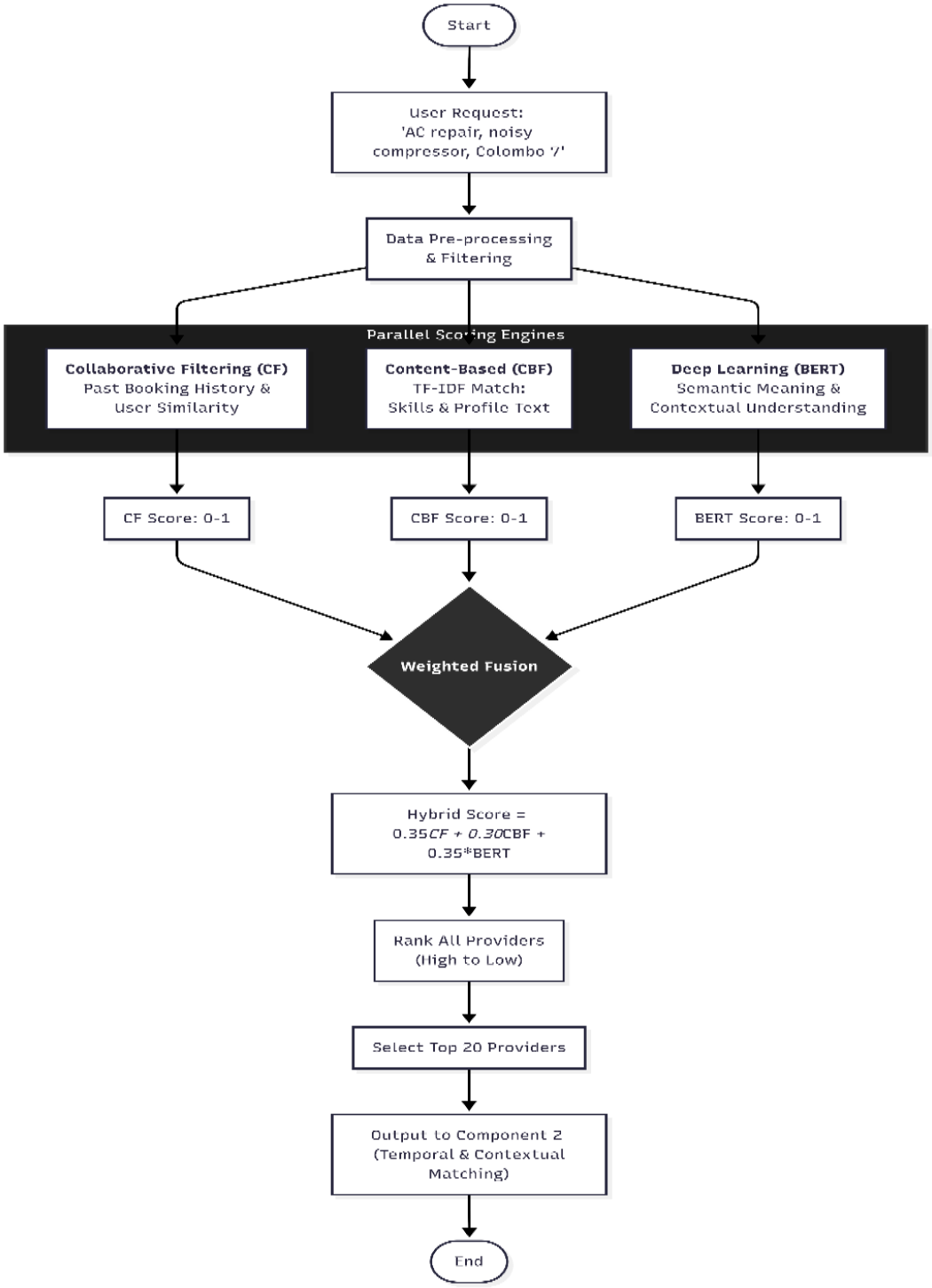


Figure 3 - Flow Chart (component 1)

4.4 Component Interactions

The system follows a pipe-and-filter architecture:

1. The API Gateway receives a request and triggers Component 1.
2. Component 1 queries the Vector Database and passes the "Top 20" to Component 2.
3. Component 2 applies spatio-temporal filters and passes the refined list to Component 3.
4. Component 3 validates credentials against the Relational Database before Component 4 performs the final sentiment ranking.

4.5 Data Flow Explanation

The data flow within the proposed platform follows a Pipe-and-Filter architectural pattern, ensuring that each component incrementally refines the data to produce high-precision results with minimal latency.

Step 1: Input Transformation and Initial Retrieval

The process begins when a consumer enters a service request (e.g., *"My AC is leaking water"*) into the web interface.

- **Semantic Vectorization:** The API Gateway sends this raw text to Component 1, where the BERT model transforms the natural language query into a high-dimensional mathematical vector (embedding).
- **Hybrid Candidate Selection:** This vector is compared against thousands of provider profiles stored in the Vector Database. Simultaneously, the Collaborative Filtering logic pulls historical interaction data from the PostgreSQL database. These scores are fused to extract the Top 20 candidates who best match the user's intent and behavioral patterns.

Step 2: Contextual Filtering and Real-Time Adjustment

The "Top 20" list is passed to Component 2. Unlike the first stage, which is based on historical and semantic data, this stage uses volatile real-time data.

- **Proximity Calculation:** Using the consumer's IP/GPS coordinates and the providers' stored locations, the system calculates distance via the Haversine formula.
- **Environmental Overlay:** Real-time weather APIs and availability status are checked. If a provider is marked "Busy" or if a weather event makes the service impossible (e.g., outdoor roofing during a storm), the provider is deprioritized or removed, resulting in a Refined List.

Step 3: Credibility and Trust Verification

The Refined List moves to Component 3 to undergo a security and quality check.

- **Credential Retrieval:** The system pulls digital certifications and past task completion rates from the PostgreSQL database.
- **Weighted Trust Score:** Each provider is assigned a Credibility Score. This is a non-static value that fluctuates based on recent performance. Only those meeting a specific trust threshold form the Verified List.

Step 4: Sentiment Analysis and Final Ranking

The final refinement happens in Component 4.

- **Review Processing:** The system retrieves the text of past customer reviews for the remaining verified providers.
- **Aspect-Level Extraction:** The LSTM model processes these reviews to find specific "aspects" like *punctuality* or *professionalism*.
- **Final Rank Generation:** These sentiment scores are used to re-order the list. The Final Top 5 providers are then packaged into a JSON response by the API Gateway and pushed back to the Web App (Consumer).

Data Integrity and Latency Control

- **Unidirectional Flow:** By moving data in one direction, the system avoids complex feedback loops that would increase response time (latency).
- **Caching:** Frequently accessed provider metadata is cached to ensure that the "real-time" adjustments in Component 2 do not cause bottlenecks.

4.6 Scalability and Security Considerations

- **Scalability:** The use of FastAPI and Docker allows individual components to be scaled independently as demand grows.
- **Security:** Data integrity is maintained through PostgreSQL security protocols, and sensitive provider credentials are encrypted to ensure privacy and ethical compliance.

5. User Requirements

5.1 Stakeholder Identification

Identifying the key participants is essential to ensure the system meets the diverse needs of the Sri Lankan service market.

- **Primary Consumer:** Individuals seeking maintenance services (e.g., plumbing, electrical, AC repair) who require quick, reliable, and semantically accurate recommendations.

- **Service Provider:** Skilled technicians and maintenance professionals who need a platform to showcase their verified skills and receive job leads based on their location and expertise.
- **System Administrator:** Personnel responsible for managing the platform, verifying provider credentials, and monitoring the ML pipeline's performance.
- **Research Supervisor/Evaluator:** Academic stakeholders who assess the system's technical novelty and its ability to meet the 88% precision target.

5.2 Functional Requirements

These requirements define the specific actions the system must be able to perform.

1. **Semantic Request Processing:** The system shall allow users to input service requests in natural language (Sinhala/English) and extract intent using BERT.
2. **Hybrid Provider Matching:** The system shall generate an initial list of 20 matches using a weighted fusion of BERT, Collaborative Filtering, and Content-Based Filtering.
3. **Real-time Geolocation Filtering:** The system shall filter providers based on their proximity to the user using the Haversine formula.
4. **Credential Verification:** The system shall allow service providers to upload certifications and cross-reference them to calculate a credibility score.
5. **Sentiment-Based Ranking:** The system shall perform aspect-level sentiment analysis on historical reviews using an LSTM model to re-rank the top candidates.
6. **Final Recommendation Display:** The system shall present a "Top 5" recommended provider list to the consumer through the web interface.
7. **Profile Management:** Service providers shall be able to update their skills, availability, and location data in real-time.

5.3 Non-Functional Requirements

These requirements define the quality attributes and constraints of the system.

- **Performance (Latency):** The end-to-end processing of the 4-stage ML pipeline must be completed in under **2.0 seconds** to ensure a seamless web experience.
- **Accuracy (Precision):** The system must achieve a recommendation precision of at least 88% during the evaluation phase.
- **Security:** All user and provider data must be encrypted using industry-standard protocols; sensitive PII must be scrubbed before being used in ML training.
- **Usability:** The web interface must be responsive and accessible on both desktop and mobile browsers, catering to users with varying levels of technical literacy.

- **Scalability:** The backend (FastAPI and PostgreSQL) must be capable of handling an increasing number of concurrent service requests without a significant drop in performance.
- **Availability:** The platform should maintain 99% uptime to ensure users can request emergency maintenance services at any time.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Understanding the "Human" side of the system is vital for the 88% precision target.

- **Target Market:** The focus is on the Urban Maintenance Sector in Sri Lanka. Currently, this market is fragmented, relying on word-of-mouth or unverified social media posts.
- **Customer Persona - The Tech-Savvy Urbanite:** This user lives in Colombo or Gampaha, uses a smartphone for everything, but fears being "ripped off" by unverified workers. They need a **Web Platform** that acts as a middleman for trust.
- **Provider Persona - The Skilled Freelancer:** Skilled individuals who lack marketing budgets. Your platform gives them a "Credibility Score" based on their actual skills, levelling the playing field against large maintenance companies.

6.2 Value Proposition

The novelty lies in the Multi-Stage Refinement.

- **Precision as a Product:** While competitors offer search, you offer matching. The value is in the "Top 5" list which saves the user from calling 10 different people to find one who is available.
- **Bilingual Intelligence:** BERT allows users to type in "Singlish" or Sinhala, which is the primary mode of communication in local maintenance, making the platform more inclusive than English-only apps.

6.3 Revenue Model

- **Transaction Fee:** A success-based fee. This ensures the platform's interests are aligned with the user (money is only made if a good match happens).
- **Data Insights (B2B):** Aggregated, anonymized data on service demand patterns (e.g., "Which areas need more plumbers?") can be sold to training institutes or tool manufacturers.

6.4 Cost Estimation

- **Infrastructure:** High-performance GPU instances (like AWS p3) for the initial training of the LSTM and BERT models.
- **Data Acquisition:** Costs associated with scraping or purchasing local service datasets to solve the "Cold Start" problem in Recommendation Engines.
- **Hosting:** Monthly costs for PostgreSQL and Vector DB (Pinecone) to maintain the 2.0s latency requirement.

6.5 Pricing Strategy

- **Freemium for Providers:** Basic listing is free; a small monthly fee for a "Verified Badge" (which requires Component 3 verification).
- **Dynamic Pricing:** In the future, the platform could suggest higher prices during high-demand periods (e.g., AC repair during heatwaves), benefiting the providers while taking a commission.

6.6 Competitive Advantage

- **Semantic Advantage:** Unlike ikman.lk, your system understands that "pipe ekak hayi karanna" means "plumbing."
- **Credibility Advantage:** Most platforms are "pay-to-play" (whoever pays more is at the top). Your system is Performance-Based, ranking people by sentiment and verified history.

6.7 Intellectual Property (IP) Considerations

- **Software Copyright:** The source code of the **FastAPI** backend and the specific **BERT-LSTM** fusion logic.
- **Database Rights:** The unique dataset of Sinhala-English service intents and the "Vector Embeddings" of local providers.
- **Patenting:** The specific 4-stage pipeline logic could be considered for a "Business Process Patent" if it shows a significant leap over existing state-of-the-art systems.

7.BUDGET AND JUSTIFICATION

7.1 Estimated Budget Breakdown

Table 5 - Summary of Estimated Budget

Category	Item Description	Estimated Cost (LKR)
Hardware	Existing Personal Computer (Laptop/Desktop)	0 (Existing)
Software	Open Source Libraries (FastAPI, PyTorch, BERT)	0 (Free)
Cloud Services	Google Colab (Free Tier) & Render/Railway Free Tier	0 (Free)
Connectivity	High-speed Internet for Model Training & Research	15,000
Software/Web	Domain Name (.com/.lk) & Basic Hosting (Annual)	8,000
Documentation	Printing, Binding, and Thesis Preparation	7,000
Total		30,000

7.2 Detailed Justification for Each Expense

7.2.1 Hardware Costs

- **Justification:** As this is an academic research project, the primary hardware used will be the our existing personal computer.

7.2.2 Software Licenses

- **Justification:** All software frameworks, including Python, FastAPI, and BERT, are Open Source, meaning there are no licensing costs involved.[19]

7.2.3 Cloud Services

- **Justification:** To train the BERT and LSTM models, Google Colab **(**Free Tier) [20] will be utilized, providing access to high-performance GPUs at no cost. For the final web application hosting, platforms with a free entry tier (e.g., Vercel) will be used to demonstrate the prototype.

7.2.4 Connectivity and Research Access

- **Justification:** Consistent high-speed internet is essential for downloading large pre-trained models (BERT-base)[19] and accessing IEEE/ACM research databases. This budget covers the specialized data requirements for the project duration.

7.2.5 Web Hosting and Domain

Justification: To evaluate the User Requirements, a live web URL is needed. A nominal fee is allocated for a professional domain and basic web hosting to make the system accessible to stakeholders for testing.

8.WORK BREAKDOWN STRUCTURE

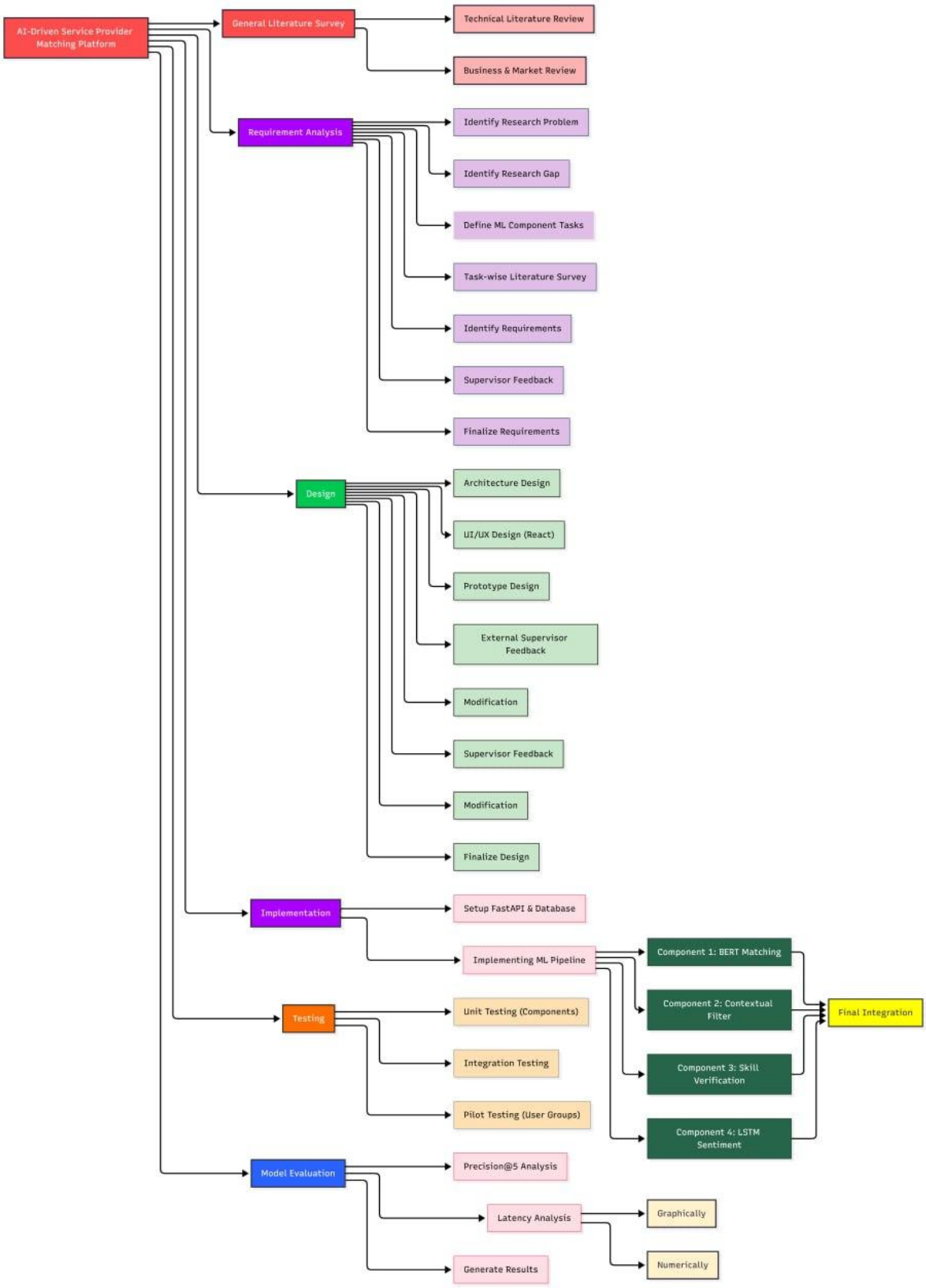


Figure 4 - Work Breakdown Structure

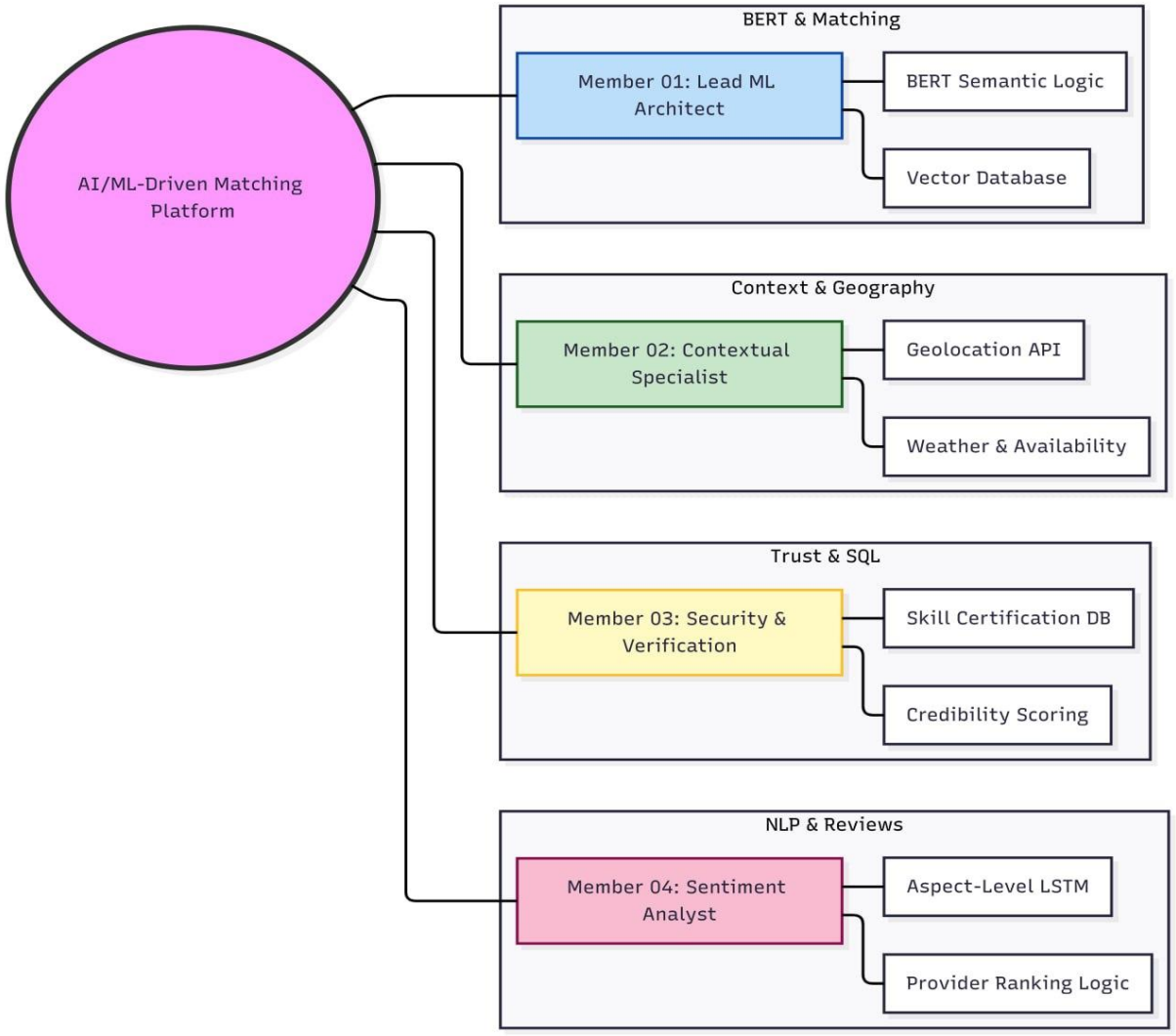


Figure 5 - Member-wise Work Breakdown Structure

9.GANTT CHART

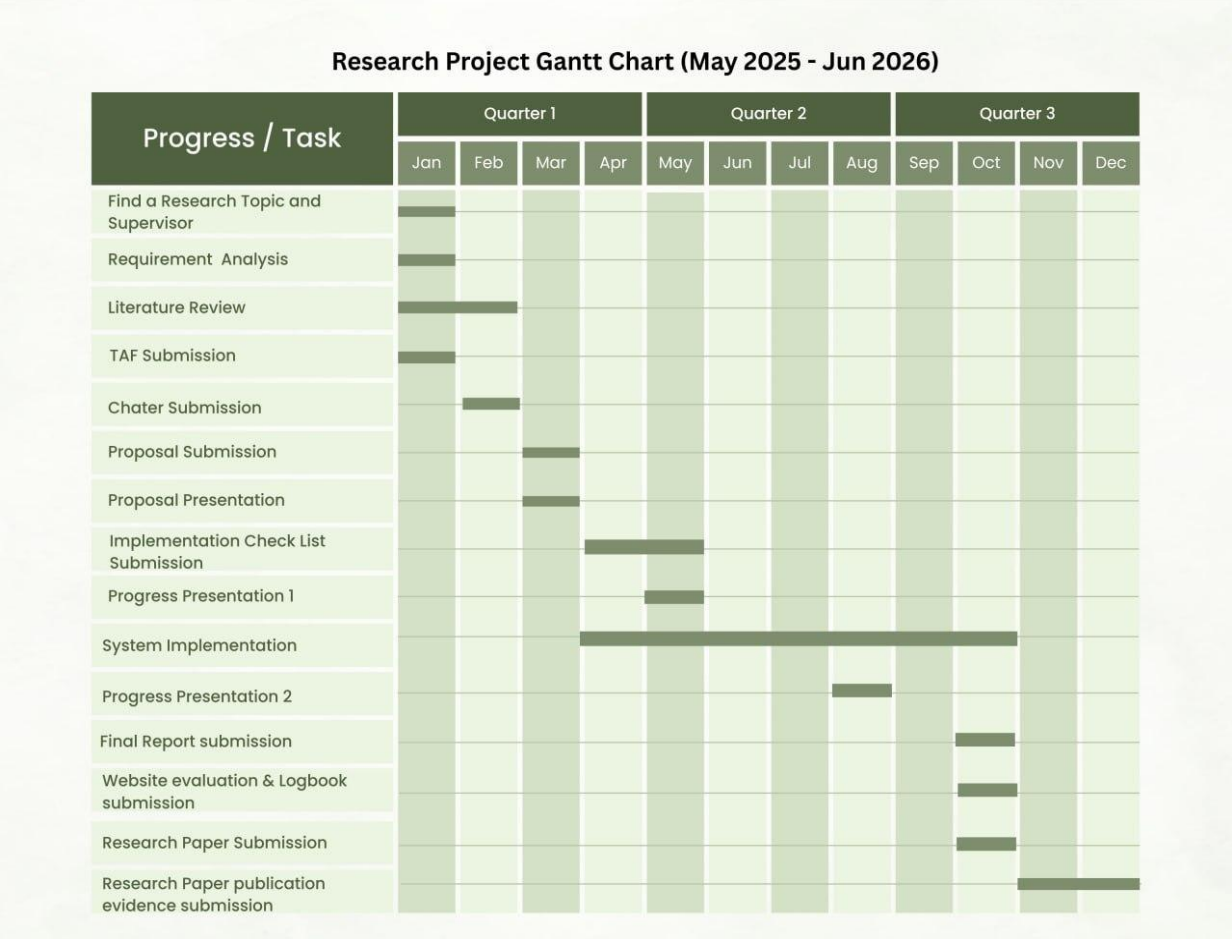


Figure 6 - Gantt Chart

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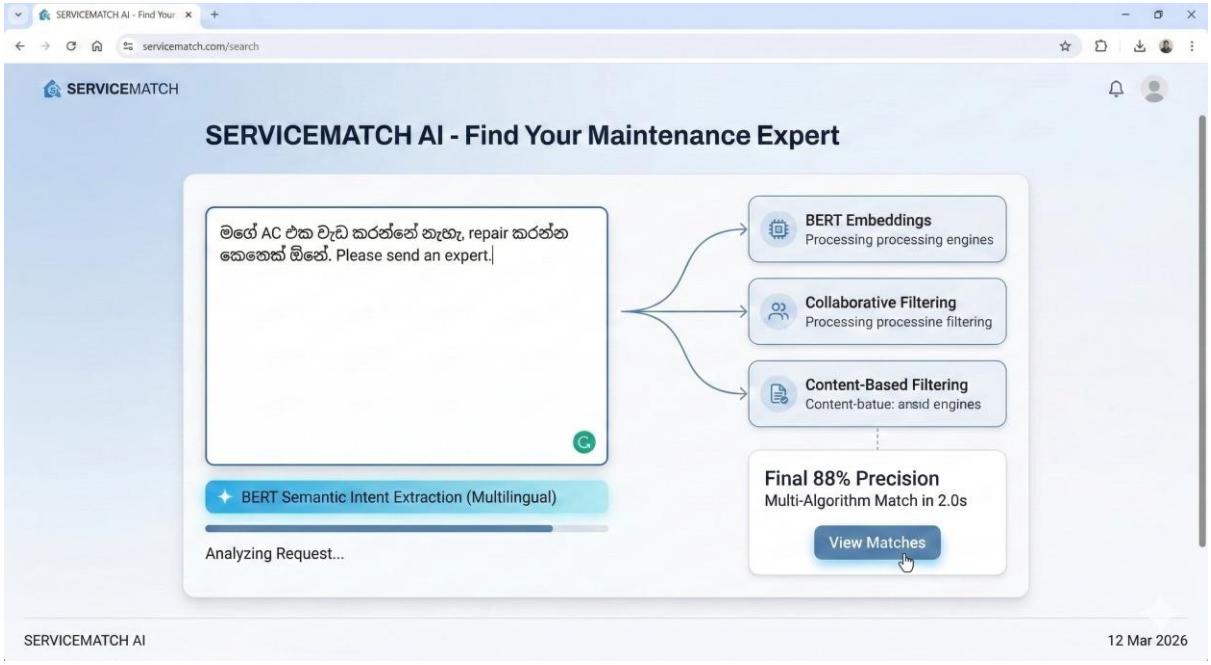
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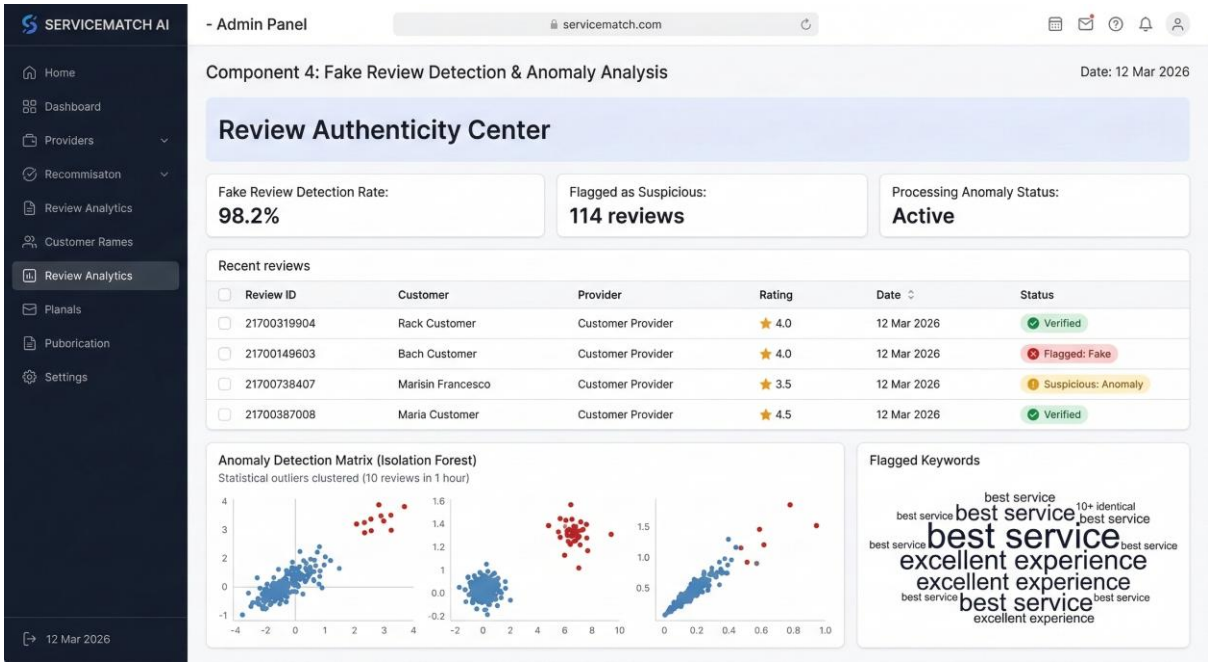
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11.APPENDICES

11.1 UI Mockups



Appendix 1



Appendix 2

11.2 Risk Assessment and Mitigation

Risk Category	Risk Description	Probability	Impact	Risk Score (0-10)	Mitigation Strategy
Technical	BERT/LSTM models fail to meet accuracy targets (85**%**) due to bilingual nuances.	Medium	High	6	Fine-tune models with a local bilingual corpus; use Attention Mechanisms to focus on key intent.
Data	Limited availability of historical booking data for Collaborative Filtering.	High	High	9	Use Hybrid Fusion to give more weight to BERT (35%) and CBF (30%) until data grows.
Security	Leakage of sensitive user data (geolocation, contact details, payment info)	Low	Very High	8	Implement AES encryption, strict access control policies, and secure database schemas.
Performance	Total pipeline latency exceeds 10 seconds per request.	Medium	Medium	4	Implement parallel processing for CF, CBF, and BERT and use caching for weather/location data.
Business / Adoption	Users or providers resist adoption due to lack of trust in AI-driven matching.	Medium	Medium	4	Provide transparent explanations for recommendations; ensure a bilingual (Sinhala/English) user-friendly UI.
External	Dependency on external Weather/Map APIs leads to system failures during outages.	Medium	High	4	Enable offline fallback logic; cache the most recent location and weather data locally for 2 hours.
Resource	Unavailability of expert technicians to verify the	Low	High	5	Collaborate with vocational training institutes (e.g., NAITA) to

	ground truth for training.				build a verified expert database.

Appendix 3